

PROJECT DATA SHEET

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JOB/PROJECT : DIPTFORD PRIMARY SCHOOL**JOB/PROJECT NO :** C8611 **SHEET NO :** COVER**GENERAL :****CLIENT :** MW BENNY CONSTRUCTION**ARCHITECT :** KAY TRAGENNA**AUTHORITY :** SOUTH HAMS COUNCIL**JOB/PROJECT DESCRIPTION :** SCHOOL EXTENSION FOUNDATION DESIGN**PROPOSED USE:** CLASSROOM**OVERALL SIZE :** AS CALCS**No. OF STOREYS :** ONE**TYPE OF SOIL INVESTIGATION :** WINDOW SAMPLING IN RED
ROCK GEO REPORT REFERENCE RP7952**BEARING STRATA :** DEVONIAN SLATE**TYPE OF FOUNDATION :** REINFORCED BEAM AND PILES**MAXIMUM SAFE BEARING PRESSURE** N/A kN/m²**STRUCTURAL MATERIALS USED :** RC32/40 CONCRETE**STANDARDS AND CODES USED IN DESIGN :**

General actions: Densities, self weight & imposed loads	BS EN 1991-1-1	☒	Steelwork	BS EN 1993 (EC3)	☐
General actions: Actions on structures exposed to fire	BS EN 1991-1-2	☐	Timber	BS EN 1995 (EC5)	☐
Snow Loads	BS EN 1991-1-3	☐	Masonry	BS EN 1996 (EC6)	☐
Wind Loads	BS EN 1991-1-4	☐	Others		☐
Concrete	BS EN 1992 (EC2)	☒			☐

AUTHORISATION OF CALCULATION SHEETS AND BENDING SCHEDULES :

CALCULATION SHEET NOS.	CHECKED BY	DATE
Sheet nos: 01 — 14 inclusive	<i>A.C. HEARD</i>	<i>25/07/23</i>



Calculations

Contract	DIPTFORD PRIMARY SCHOOL TOTNESS	Job No.	C8611	Sheet No.	01
		Designed by	D.P.H	Date:	JULY 23

DESCRIPTION - For layout refer to sheet 14

These calculations cover the design of the substructures for this proposed classroom extension. All information has been taken from Kay Tragenna Architects drawings.

The loadings from the superstructure have been verified against the timber frame design undertaken by Allwood timber

The ground investigation by Red Rock has established that a piled foundation solution is required to limit differential settlement to acceptable limits. These calculations establish the required loads to be used for the pile design.

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Calculations

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LOADING

FLAT ROOF

$$\begin{aligned}\text{Single ply membrane} &= 0.10 \text{ kN/m}^2 \\ \text{Insulation} &= 0.05 \text{ kN/m}^2 \\ \text{Plywood} &= 0.15 \text{ kN/m}^2 \\ \text{Structural selfweight} &= 0.20 \text{ kN/m}^2 \\ \text{Suspended ceiling} &= 0.50 \text{ kN/m}^2 \\ \Sigma &= 1.00 \text{ kN/m}^2 \times 1.35 = 1.35 \text{ kN/m}^2 \\ \text{Lh} &= 0.60 \text{ kN/m}^2 \times 1.50 = 0.90 \text{ kN/m}^2 \\ \text{P.V. Allowance} &= 0.25 \times 0.7 = 0.18 \text{ kN/m}^2 \times 1.50 = 0.27 \text{ kN/m}^2 \\ \text{Services} &= 0.25 \text{ kN/m}^2 \times 1.50 = 0.38 \text{ kN/m}^2 \\ \Sigma &= 2.03 \text{ kN/m}^2 \quad \Sigma = 2.90 \text{ kN/m}^2\end{aligned}$$

ELEVATION WALL

$$\begin{aligned}\text{Wood cladding} &= 0.20 \text{ kN/m}^2 \\ \text{Battens} &= 0.05 \text{ kN/m}^2 \\ \text{OSB} &= 0.10 \text{ kN/m}^2 \\ \text{Studs} &= 0.15 \text{ kN/m}^2 \\ \text{Plasterboard} &= 0.25 \text{ kN/m}^2 \\ \Sigma &= 0.75 \text{ kN/m}^2 \times 1.35 = 1.01 \text{ kN/m}^2\end{aligned}$$



Calculations

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		Designed by	<u>D.P.H</u>	Date:	<u>JULY 23</u>

INTERNAL WALL

$$\begin{aligned}\text{Plasterboard} &= 0.50 \text{ kN/m}^2 \\ \text{Structural selfweight} &= 0.15 \text{ kN/m}^2 \\ \text{OSB} &= 0.10 \text{ kN/m}^2 \\ \Sigma &= 0.75 \text{ kN/m}^2 \times 1.35 = 1.01 \text{ kN/m}^2\end{aligned}$$

FLOOR

$$\begin{aligned}\text{Beam \& Block} &= 2.75 \text{ kN/m}^2 \\ \text{Insulation} &= 0.05 \text{ kN/m}^2 \\ \text{Screed} &= 1.80 \text{ kN/m}^2 \\ \Sigma &= 4.60 \text{ kN/m}^2 \times 1.35 = 6.21 \text{ kN/m}^2 \\ \text{Lh} &= 3.00 \text{ kN/m}^2 \times 1.50 = 4.50 \text{ kN/m}^2 \\ \text{Partition Allowance} &= 0.50 \text{ kN/m}^2 \times 1.50 = 0.75 \text{ kN/m}^2 \\ \Sigma &= 8.10 \text{ kN/m}^2 \quad \Sigma = 11.46 \text{ kN/m}^2\end{aligned}$$



Calculations

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CAPACITY CHECK OF CONCRETE BEAMS

- All Beams are 450×450 with 4No. H16 bars T&B and 2No. H10 Links in pairs @ 200 c/s.

Moment capacity check:

$$d_{eff} = 450 - 50 - 10 - \frac{16}{2} = 382 \text{ mm}$$

↓ ↓ ↓
Cover Shearlinks half bar diameter

$$Z = 0.95 \times d_{eff} = 0.95 \times 382 = 360 \text{ mm}$$

$$A_{st} = 4 \times (\pi \times \frac{16^2}{4}) = 800 \text{ mm}^2$$

$$\therefore M_R = 800 \times 500 \times 0.87 \times 360 = 125 \text{ kNm}$$

$$k = \frac{M}{bd^2 f_{cu}} = \frac{125 \times 10^6}{450 \times 382^2 \times 32} = 0.059 \quad \therefore \frac{Z}{d} = 0.95 \quad \therefore M_R = 125 \text{ kNm}$$

Shear resistance check:

$$V_{min} = 0.15 \times 450 \times 382 \times \sqrt{32} = 146 \text{ kN}$$

$$A_{sw} = \frac{S \times 0.08 \times b \times \sqrt{f_{cu}}}{f_{ys}} = \frac{200 \times 0.08 \times 450 \times \sqrt{32}}{500} = 81.5 \text{ mm}^2$$

$$A_s \text{ Provided} = 4 \times (\pi \times \frac{10^2}{4}) = 314 \quad \therefore \text{OK}$$

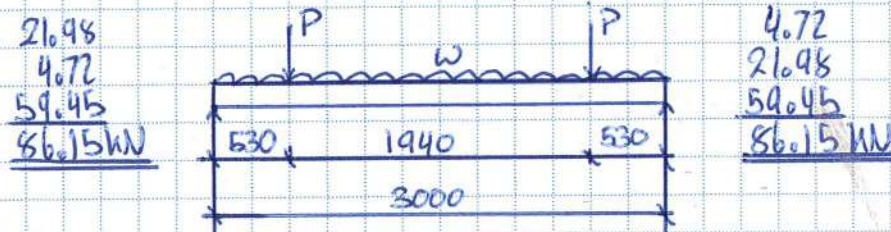
$\therefore \text{MAX MOMENT} = 125 \text{ kNm} \quad \& \quad \text{MAX SHEAR} = 146 \text{ kN}$



Calculations

Contract	DIPTFORD PRIMARY SCHOOL TOTNESS	Job No.	CS611	Sheet No.	05
		Designed by	D.P.O.H	Date:	JULY 23

GROUND BEAM 1 - GRID A 7-8



$$P = 12 \times 1.35 = 16.2 \text{ kN}$$

$$7 \times 1.50 = 10.5 \text{ kN}$$

$$\underline{26.7 \text{ kN}}$$

(From Allwood line & pointload sketch)

$$W = \begin{aligned} \text{Roof} &= (0.48 \times 2) \times 2.90 = 0.70 \text{ kN/m} \\ \text{Wall} &= 3.0 \times 1.01 = 3.03 \text{ kN/m} \\ \text{Floor} &= (4.088 \div 2) \times 11.46 = 23.42 \text{ kN/m} \\ \text{Wall} &= 0.675 \times 0.325 \times 20 \times 1.35 = 5.92 \text{ kN/m} \\ \text{Beam self weight} &= 0.45^3 \times 24 \times 1.35 = 6.56 \text{ kN/m} \end{aligned}$$

$$\underline{Z = 39.63 \text{ kN/m}}$$

$$M_d = (86.15 \times 1.50) - (26.7 \times 0.47) - (39.63 \times \frac{1.5^2}{2}) = 58.7 \text{ kNm}$$

$$58.7 < 125 \quad \& \quad 86.2 < 146 \quad \therefore \text{OK}$$

GROUND BEAM 1 - GRID A 8-9

$$W = \begin{aligned} \text{Roof} &= \frac{0.48}{2} \times 2.90 = 0.70 \text{ kN/m} \\ \text{Wall} &= 3.0 \times 1.01 = 3.03 \text{ kN/m} \\ \text{Floor} &= \frac{4.088}{2} \times 11.46 = 23.42 \text{ kN/m} \\ \text{Wall} &= 0.675 \times 0.325 \times 20 \times 1.35 = 5.92 \text{ kN/m} \\ \text{Beam self weight} &= 0.45^3 \times 24 \times 1.35 = 6.56 \text{ kN/m} \end{aligned}$$

$$\underline{Z = 39.63 \text{ kN/m}}$$

$$\text{Design Span} = 3.44 \text{ m}$$

$$M_d = 39.63 \times \frac{3.44^2}{8} = 58.62 \text{ kNm} \quad V_A = V_B = 68.16 \text{ kN}$$

$$58.62 < 125 \quad \& \quad 68.16 < 146 \quad \therefore \text{OK}$$



Calculations

Contract	DIPTFORD PRIMARY SCHOOL TORNESS	Job No.	C8611	Sheet No.	06
		Designed by	D.P.H	Date:	JULY 23

GROUND BEAM 2 - GRID B 8-9

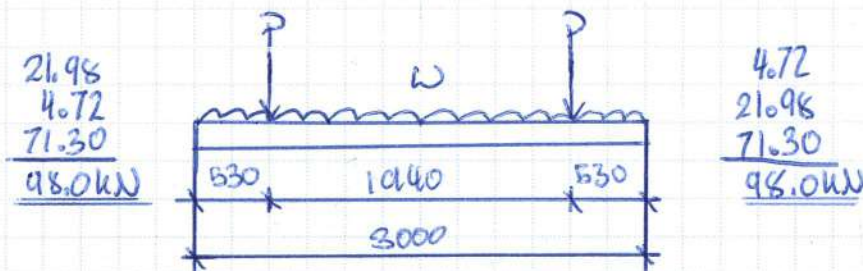
$$\begin{aligned}
 w &= \text{Roof} = 0 &= 0 \text{ kN/m} \\
 &\text{Wall} = 0 &= 0 \text{ kN/m} \\
 &\text{Floor} = \frac{4.088 + 4.087}{2} \times 11.46 &= 46.84 \text{ kN/m} \\
 &\text{Wall} = 1.125 \times 0.215 \times 20 \times 1.35 &= 6.53 \text{ kN/m} \\
 \text{Beam self weight} &= 0.45^2 \times 24 \times 1.35 &= 6.56 \text{ kN/m} \\
 \Sigma &= \underline{\underline{59.93 \text{ kN/m}}}
 \end{aligned}$$

Design Span = 3.44m

$$M_d = 59.93 \times \frac{3.44^2}{8} = 89 \text{ kNm} \quad V_A = V_B = 103 \text{ kN}$$

$$89 < 125 \quad \& \quad 103 < 146 \quad \therefore \text{OK.}$$

GROUND BEAM 3 GRID. C 7-8



$$\begin{aligned}
 P &= 12 \times 1.35 = 16.2 \text{ kN} \\
 7 \times 1.50 &= 10.5 \text{ kN} \quad (\text{From Allwood line \& point load sketch}) \\
 \Sigma &= \underline{\underline{26.7 \text{ kN}}}
 \end{aligned}$$

$$\begin{aligned}
 w &= \text{Roof} = (0.48 \div 2) \times 2.90 &= 0.70 \text{ kN/m} \\
 &\text{Wall} = 3.0 \times 1.01 &= 3.03 \text{ kN/m} \\
 &\text{Floor} = (4.087 \div 2) \times 11.46 &= 23.42 \text{ kN/m} \\
 &\text{Wall} = 1.575 \times 0.325 \times 20 \times 1.35 &= 13.82 \text{ kN/m} \\
 \text{Beam self wgt.} &= 0.45^2 \times 24 \times 1.35 &= 6.56 \text{ kN/m} \\
 \Sigma &= \underline{\underline{47.53 \text{ kN/m}}}
 \end{aligned}$$

$$M_d = (98.0 \times 1.50) - (26.7 \times 0.97) - (47.53 \times \frac{1.5^2}{2}) = 67.6 \text{ kNm}$$

$$67.6 < 125 \quad \& \quad 98 < 146 \quad \therefore \text{OK.}$$

**Calculations**

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	<u>TOTNESS</u>	Designed by	<u>D.P.H</u>	Date:	<u>JULY 23</u>

GROUND BEAM 3 - GRID C 8-9

$$w = \text{Roof} = \frac{0.48}{2} \times 2.90 = 0.70 \text{ kN/m}$$

$$\text{Wall} = 3.0 \times 1.01 = 3.03 \text{ kN/m}$$

$$\text{Floor} = \frac{4.087}{2} \times 11.46 = 23.42 \text{ kN/m}$$

$$\text{Wall} = 1.575 \times 0.325 \times 20 \times 1.35 = 13.82 \text{ kN/m}$$

$$\text{Beam self weight} = 0.45^2 \times 24 \times 1.35 = 6.56 \text{ kN/m}$$

$$\Sigma = \underline{\underline{47.53 \text{ kN/m}}}$$

$$\text{Design Span} = 3.044 \text{ m}$$

$$M_d = 47.53 \times \frac{3.044^2}{8} = 70.31 \text{ kNm} \quad V_A = V_B = 81.8 \text{ kN}$$

$$70.31 < 125 \quad \& \quad 81.8 < 146 \quad \therefore \text{OK.}$$

GROUND BEAM 4 - GRID 1 A-B

$$w = \text{Roof} = \frac{4.410}{2} \times 2.90 = 6.39 \text{ kN/m}$$

$$\text{Wall} = 3.0 \times 1.01 = 3.03 \text{ kN/m}$$

$$\text{Floor} = 0 = 0 \text{ kN/m}$$

$$\text{Wall} = 1.575 \times 0.325 \times 20 \times 1.35 = 13.82 \text{ kN/m}$$

$$\text{Beam self weight} = 0.45^2 \times 24 \times 1.35 = 6.56 \text{ kN/m}$$

$$\Sigma = \underline{\underline{29.80 \text{ kN/m}}}$$

$$\text{Design Span} = 4.09 \text{ m}$$

$$M_d = 29.80 \times \frac{4.09^2}{8} = 62.31 \text{ kNm} \quad V_A = V_B = 60.94 \text{ kN}$$

$$62.3 < 125 \quad \& \quad 60.9 < 146 \quad \therefore \text{OK.}$$



Calculations

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GROUND BEAM 5. - GRID 3 up to A

$$w = \text{Roof} = \frac{2.849 + 2.562}{2} \times 2.90 = 7.89 \text{ kN/m}$$

$$w_{\text{Wall}} = 2.70 \times 1.01 = 2.73 \text{ kN/m}$$

$$w_{\text{Floor}} = \frac{2.849}{2} \times 11.46 = 16.32 \text{ kN/m}$$

$$w_{\text{Wall}} = 0.225 \times 0.325 \times 20 \times 1.35 = 1.97 \text{ kN/m}$$

$$\text{Beam self weight} = 0.45^2 \times 24 \times 1.35 = 6.56 \text{ kN/m}$$

$$\Sigma = 35.47 \text{ kN/m}$$

Design Span = 4.21m

$$M_d = 35.47 \times \frac{4.21^2}{8} = 78.58 \text{ kNm} \quad V_A = V_B = 74.66 \text{ kN}$$

Consider 1m cantilever: $M_d = \frac{35.47 \times 1.00^2}{2} = 17.74 \text{ kNm}$

$$78.58 < 125 \quad \& \quad 74.66 < 146 \quad \therefore \text{Ok}$$

GROUND BEAM 6 - GRID 4 up to A

$$w = \text{Roof} = 0 = 0 \text{ kN/m}$$

$$w_{\text{Wall}} = 0 = 0 \text{ kN/m}$$

$$w_{\text{Floor}} = \frac{2.849 + 2.562}{2} \times 11.46 = 31.01 \text{ kN/m}$$

$$w_{\text{Wall}} = 0.225 \times 0.215 \times 20 \times 1.35 = 1.31 \text{ kN/m}$$

$$\text{Beam self weight} = 0.45^2 \times 24 \times 1.35 = 6.56 \text{ kN/m}$$

$$\Sigma = 38.88 \text{ kN/m}$$

Design Span = 4.21m

$$M_d = 38.88 \times \frac{4.21^2}{8} = 86.14 \text{ kNm} \quad V_A = V_B = 81.84 \text{ kN}$$

Consider 1m cantilever: $M_d = \frac{38.88 \times 1.00^2}{2} = 19.44 \text{ kNm}$

$$86.14 < 125 \quad \& \quad 81.84 < 146 \quad \therefore \text{Ok}$$



Calculations

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		Designed by	D.P.H	Date:	JULY 23

GROUND BEAM 7

GRID 5 up to A

$$w = \text{Roof} = \frac{2.849 + 2.562}{2} \times 2.90 = 7.89 \text{ kN/m}$$

$$\text{Wall} = 2.70 \times 1.01 = 2.73 \text{ kN/m}$$

$$\text{Floor} = \frac{2.562}{2} \times 11.46 = 14.68 \text{ kN/m}$$

$$\text{Wall} = 0.225 \times 0.325 \times 20 \times 1.35 = 1.97 \text{ kN/m}$$

$$\text{Beam self weight} = 0.45^2 \times 24 \times 1.35 = 6.56 \text{ kN/m}$$

$$\Sigma = \underline{\underline{33.83 \text{ kN/m}}}$$

Design Span = 2.17m

$$M_d = 33.83 \times \frac{2.17^2}{8} = 19.91 \text{ kNm} \quad V_A = V_B = 36.71 \text{ kN}$$

$$\text{Consider 1m cantilever: } M_d = 33.83 \times \frac{1.00^2}{2} = 16.92 \text{ kNm}$$

$$19.91 < 125 \quad \& \quad 36.71 < 146 \quad \therefore \text{OK}$$

GROUND BEAM 8

GRID 6 A-B

$$w = \text{Roof} = \frac{5.083 + 4.894}{2} \times 2.90 = 14.47 \text{ kN/m}$$

$$\text{Wall} = 3.0 \times 1.01 = 3.03 \text{ kN/m}$$

$$\text{Floor} = 0 = 0 \text{ kN/m}$$

$$\text{Wall} = 0.675 \times 0.215 \times 20 \times 1.35 = 3.92 \text{ kN/m}$$

$$\text{Beam self weight} = 0.45^2 \times 24 \times 1.35 = 6.56 \text{ kN/m}$$

$$\Sigma = \underline{\underline{27.98 \text{ kN/m}}}$$

Design Span = 4.09m

$$M_d = 27.98 \times \frac{4.09^2}{8} = 58.51 \text{ kNm} \quad V_A = V_B = 57.22 \text{ kN}$$

$$58.51 < 125 \quad \& \quad 57.22 < 146 \quad \therefore \text{OK}$$

**Calculations**

Contract	DIPTFORD PRIMARY SCHOOL TOTNESS	Job No.	CB611	Sheet No.	10
		Designed by	D.P.H	Date:	JULY 23.

GROUND BEAM 9 GRID 9 A-B

$$W = \text{Roof} = \frac{4.894}{2} \times 2.90 = 7.10 \text{ kN/m}$$

$$\text{Wall} = 3.0 \times 1.01 = 3.03 \text{ kN/m}$$

$$\text{Floor} = 0 = 0 \text{ kN/m}$$

$$\text{Wall} = 0.675 \times 0.325 \times 20 \times 1.35 = 5.92 \text{ kN/m}$$

$$\text{Beam self weight} = 0.45^2 \times 24 \times 1.35 = 6.56 \text{ kN/m}$$

$$\underline{22.61 \text{ kN/m}}$$

$$\text{Design Span} = 4.09 \text{ m}$$

$$M_d = 22.61 \times \frac{4.09^2}{8} = 47.28 \text{ kNm} \quad V_A = V_B = 46.24 \text{ kN}$$

$$47.28 < 125 \quad \& \quad 46.24 < 146 \quad \therefore \text{OK}$$

CONCRETE BEAM TIES (10,11,12)

$$W = \text{Roof} = 0 = 0 \text{ kN/m}$$

$$\text{Wall} = 0 = 0 \text{ kN/m}$$

$$\text{Floor} = 0 = 0 \text{ kN/m}$$

$$\text{Wall} = 0 = 0 \text{ kN/m}$$

$$\text{Beam self weight} = 0.3^2 \times 24 \times 1.35 = 2.92 \text{ kN/m}$$

$$\underline{\underline{\Sigma = 2.92 \text{ kN/m}}}$$

$$\text{Worse case span} = 4.09 \text{ m}$$

$$M_d = 2.92 \times \frac{4.09^2}{8} = 6.11 \text{ kNm} \quad V_A = V_B = 5.97 \text{ kN}$$

$$\therefore \text{OK}$$



Calculations

Contract	DIPTFORD PRIMARY SCHOOL TOTNESS	Job No.	C8611	Sheet No.	1.1
		Designed by	DcPH	Date:	JULY 23

PILE No.	ULTIMATE LOAD	CHARACTERISTIC LOAD	DESIGN
01	$35 + 75 + 4 = 114$	$114 \div 1.4 = 81$	100
02	$39 + 84 + 4 = 127$	$127 \div 1.4 = 91$	100
03	$34 + 37 + 4 = 75$	$75 \div 1.4 = 54$	100
04	$59 + 61 = 120$	$120 \div 1.4 = 86$	100
05	$(59 + 73) \div 2 = 66$	$66 \div 1.4 = 47$	100
06	$(59 + 86) \div 2 = 73$	$73 \div 1.4 = 52$	100
07	$86 + 75 + 56 = 217$	$217 \div 1.4 = 155$	150
08	$56 + 84 + 51 + 57 = 248$	$248 \div 1.4 = 177$	200
09	$51 + 37 + 50 = 138$	$138 \div 1.4 = 99$	100
10	$50 + 67 + 57 = 174$	$174 \div 1.4 = 124$	150
11	$67 + 86 + 6 = 159$	$159 \div 1.4 = 114$	150
12	$86 + 68 + 6 = 160$	$160 \div 1.4 = 114$	150
13	$68 + 46 = 114$	$114 \div 1.4 = 81$	100
14	$61 + 90 + 61 = 212$	$212 \div 1.4 = 151$	200
15	$(90 + 90) \div 2 = 90$	$90 \div 1.4 = 64$	100
16	$(90 + 90) \div 2 = 90$	$90 \div 1.4 = 64$	100
17	$(90 + 85) \div 2 = 88$	$88 \div 1.4 = 63$	100
18	$(90 + 85) \div 2 = 88$	$88 \div 1.4 = 63$	100
19	$85 + 57 + 77 + 57 = 276$	$276 \div 1.4 = 197$	200
20	$77 + 76 + 6 = 159$	$159 \div 1.4 = 114$	150



Calculations

Contract	DIPTFORD PRIMARY SCHOOL TOTNESS	Job No.	C8611	Sheet No.	12
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21	$76 + 57 + 97 + 57 = 287$	$287 \div 1.4 = 205$	200
22	$47 + 6 + 86 + 6 = 145$	$145 \div 1.4 = 139$	150
23	$86 + 6 + 99 + 6 = 197$	$197 \div 1.4 = 141$	150
24	$99 + 46 + 46 = 191$	$191 \div 1.4 = 136$	150
25	$61 + 71 = 132$	$132 \div 1.4 = 94$	100
26	$(71 + 92) \div 2 = 82$	$82 \div 1.4 = 59$	100
27	$(71 + 92) \div 2 = 82$	$82 \div 1.4 = 59$	100
28	$(92 + 68) \div 2 = 80$	$80 \div 1.4 = 57$	100
29	$(92 + 68) \div 2 = 80$	$80 \div 1.4 = 57$	100
30	$68 + 57 + 61 = 186$	$186 \div 1.4 = 133$	150
31	$61 + 6 + 60 = 127$	$127 \div 1.4 = 91$	100
32	$60 + 57 + 80 = 197$	$197 \div 1.4 = 140$	150
33	$80 + 6 + 92 = 178$	$178 \div 1.4 = 127$	150
34	$92 + 6 + 75 = 173$	$173 \div 1.4 = 124$	150
35	$75 + 46 = 121$	$121 \div 1.4 = 86$	100



Calculations

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		Designed by	D.P.H	Date:	JULY 23

6 - Pile Geotechnical Capacity Calculation

The pile capacity calculations are presented in the tables below.

Pile Capacity Calculation

Diameter	0.3	m
A _{base}	0.07	m ²
A _{shaft}	0.94	m ²
N _c	9	
cu Profile	1	
cu @ z=0	90	250
δ(c _u)/δz	80	0 kN/m ² /m
z=0 @	-2.0	-4.0 m OD
Factor, F	1	
f _{cu}	35	N/mm ²
Structural Capacity	692.72	kN

Pile Lgth m	RL m OD	z m	γ _b kN/m ³	F x c _u kN/m ²	α	α x c _u kN/m ²	Area m ²	Ultimate Shaft Friction		Ultimate Base Capacity	Allowable Loads
								Undrained	Undrained	Undrained	Q _{ult} /3
								δ Q _{s(ult)} kN	Σ (δ Q _{s(ult)}) kN	Q _{b(ult)} kN	Undrained kN
2.0	-2.0	0.0	18	90	0.5	23					19
2.5	-2.5	0.5	18	130	0.5	55	0.47	26	26	83	36
3.0	-3.0	1.0	18	170	0.5	75	0.47	35	61	108	56
3.5	-3.5	1.5	18	210	0.5	95	0.47	45	106	134	80
4.0	-4.0	2.0	18	250	0.5	115	0.47	54	160	159	106
4.5	-4.5	2.5	18	250	0.5	125	0.47	59	219	159	126
5.0	-5.0	3.0	18	250	0.5	125	0.47	59	278	159	146
5.5	-5.5	3.5	18	250	0.5	125	0.47	59	337	159	165
6.0	-6.0	4.0	18	250	0.5	125	0.47	59	396	159	185
6.5	-6.5	4.5	18	250	0.5	125	0.47	59	455	159	205
7.0	-7.0	5.0	18	250	0.5	125	0.47	59	514	159	224
7.5	-7.5	5.5	18	250	0.5	125	0.47	59	573	159	244
8.0	-8.0	6.0	18	250	0.5	125	0.47	59	631	159	264
8.5	-8.5	6.5	18	250	0.5	125	0.47	59	690	159	283
9.0	-9.0	7.0	18	250	0.5	125	0.47	59	749	159	303
9.5	-9.5	7.5	18	250	0.5	125	0.47	59	808	159	322
10.0	-10.0	8.0	18	250	0.5	125	0.47	59	867	159	342

From specialist pile design information with 60m long piles ultimate load capacity = 396 + 159 = 555 kN.

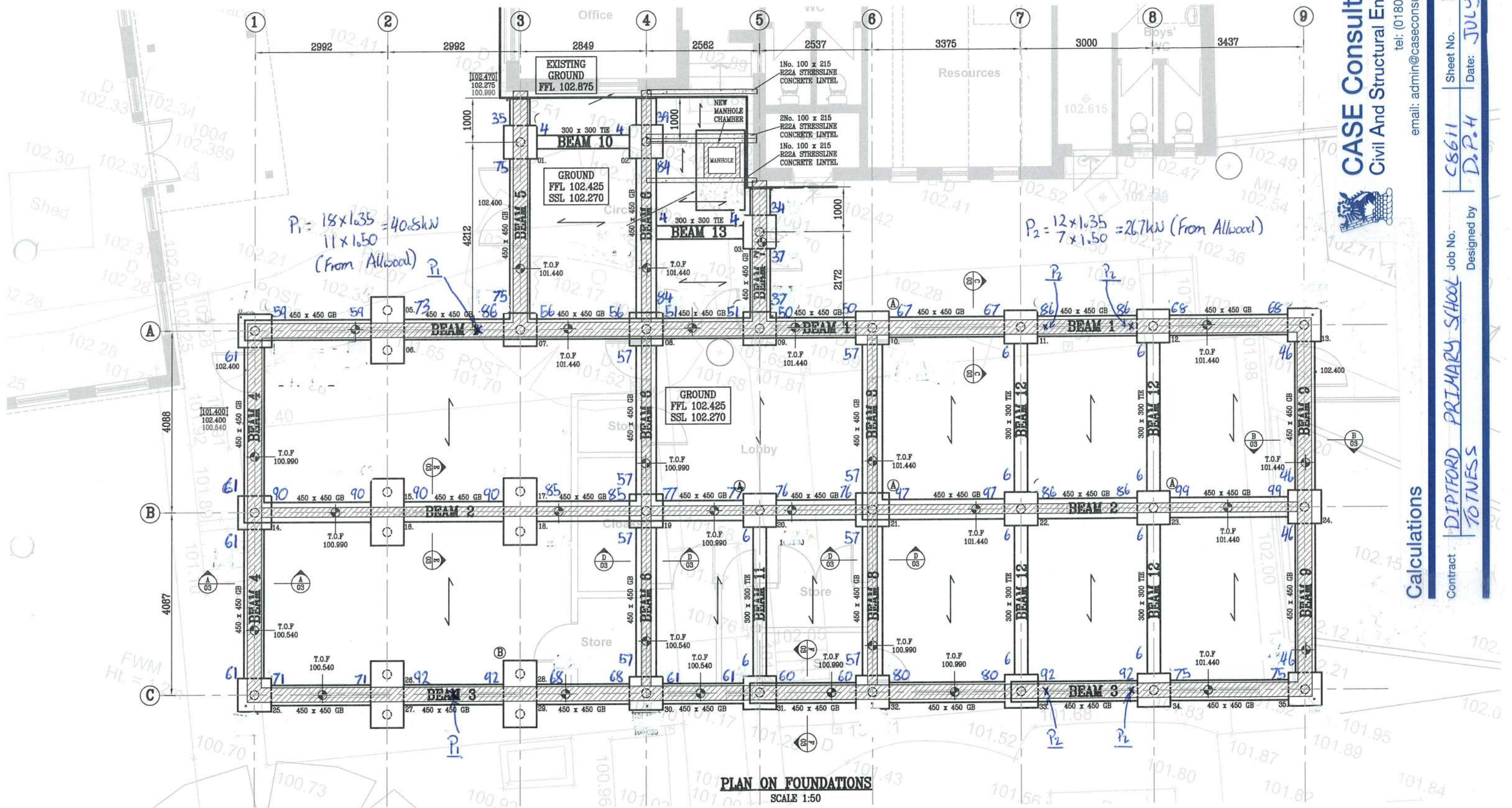
with F.O.S of 2.5 (as project drawings).

$$SWL = 555 / 2.5 = \underline{\underline{222 \text{ kN}}} \quad \text{ok.}$$

GROUND BEAM SUPPORT REACTIONS

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Contract DIPTFORD PRIMARY SCHOOL Job No. C8611 Sheet No. 14
Designed by D.P.4 Date: JULY 23



Calculations